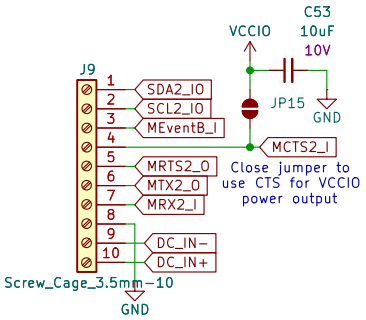
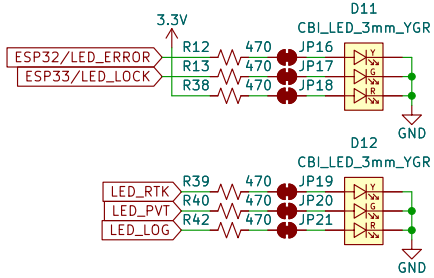
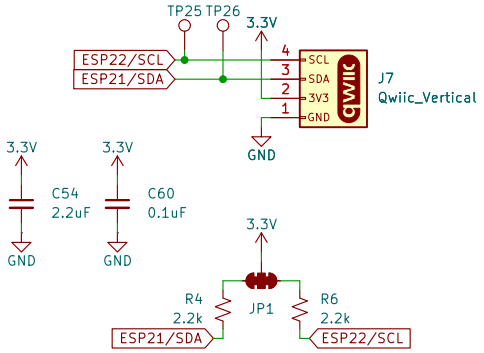
I/O Connector



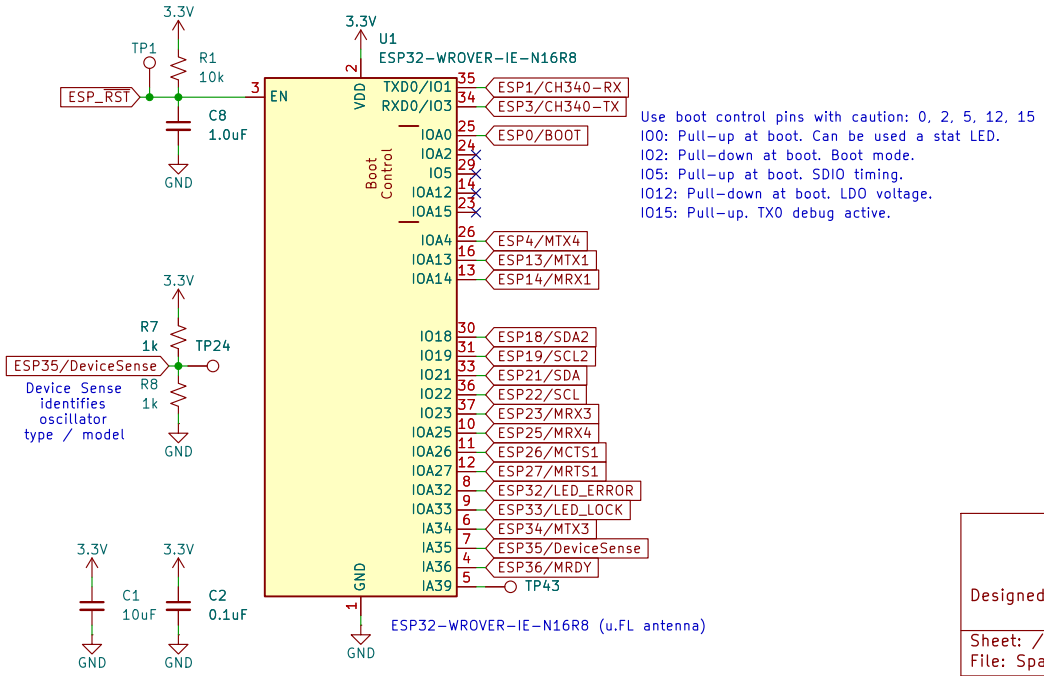
LEDs



Qwiic I²C (for OLED)



ESP32-WROVER



Power
File: Power.kicad_sch

USB
File: USB.kicad_sch

GNSS
File: GNSS.kicad_sch

Ethernet
File: Ethernet.kicad_sch

Level_Shifting
File: Level_Shifting.kicad_sch

Level_Shifting_10MHz
File: Level_Shifting_10MHz.kicad_sch

open source hardware

Designed by: P.C.

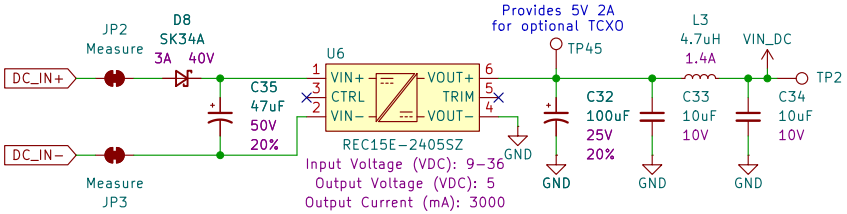
SPARKPNT

Sheet: /
File: SparkPNT_GPSDO.kicad_sch

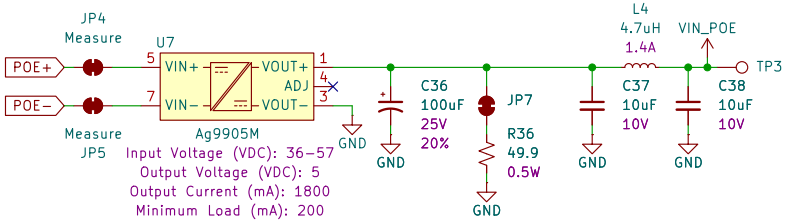
Title: GNSSDO (mosaic-T, SiT5358)

Size: USLetter	Date: 2026-08-20	**Rev: v10**
KiCad E.D.A. 10.0.1		Id: 1/7

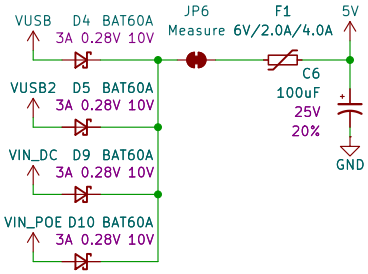
DC Power In



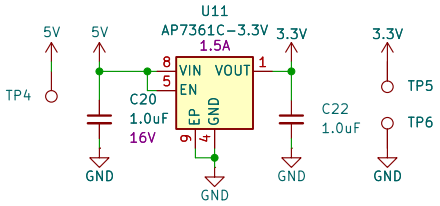
Power Over Ethernet



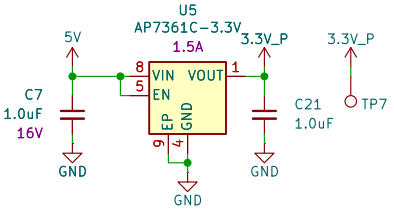
Power Mux



Main 3.3V

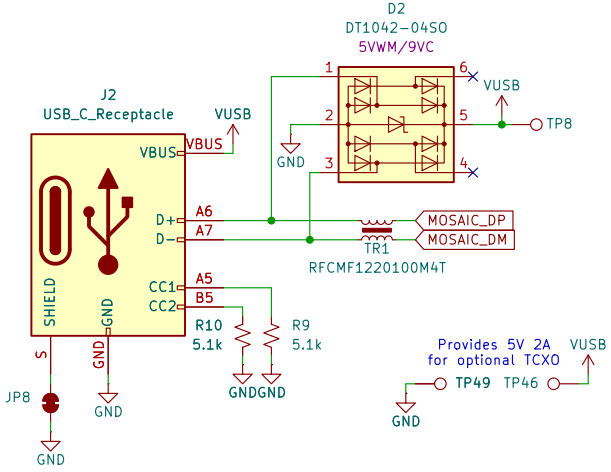


Peripheral 3.3V



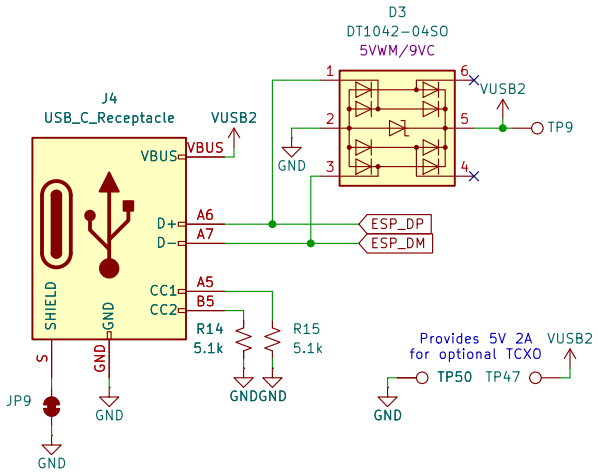
Sheet: /Power/	
File: Power.kicad_sch	
Title: Power	
Size: USLetter | Date:	**Rev:**
KiCad E.D.A. 10.0.1	Id: 2/7

Mosaic USB

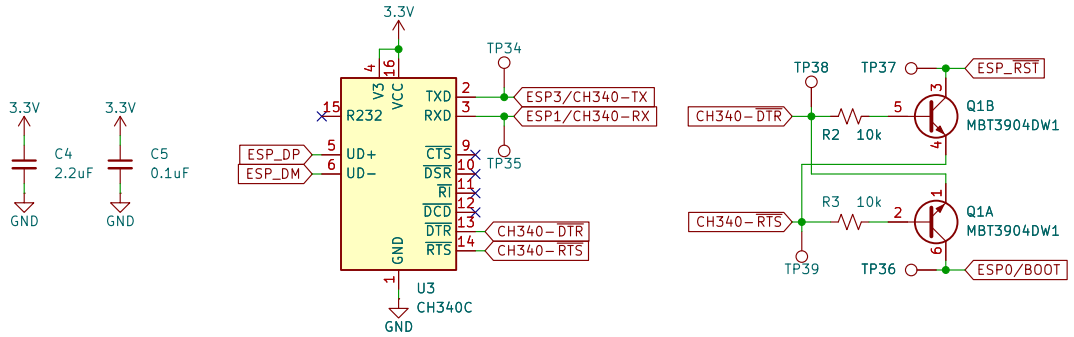


USB Track Impedance: Differential Pair
https://saturnpcb.com/saturn-pcb-toolkit/
Prepreg thickness: 8.3 mil (JLC7628). Er = 4.6
10.5 mil track with 9.5 mil gap (20 mil center to center) = 90 Ohms

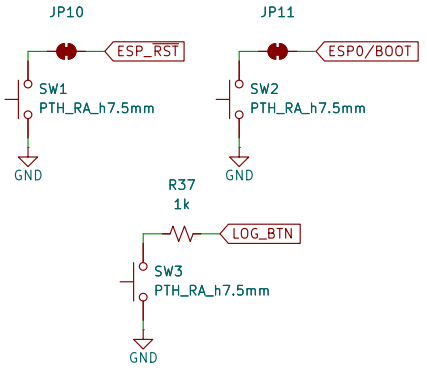
ESP32 USB



ESP32 USB to Serial

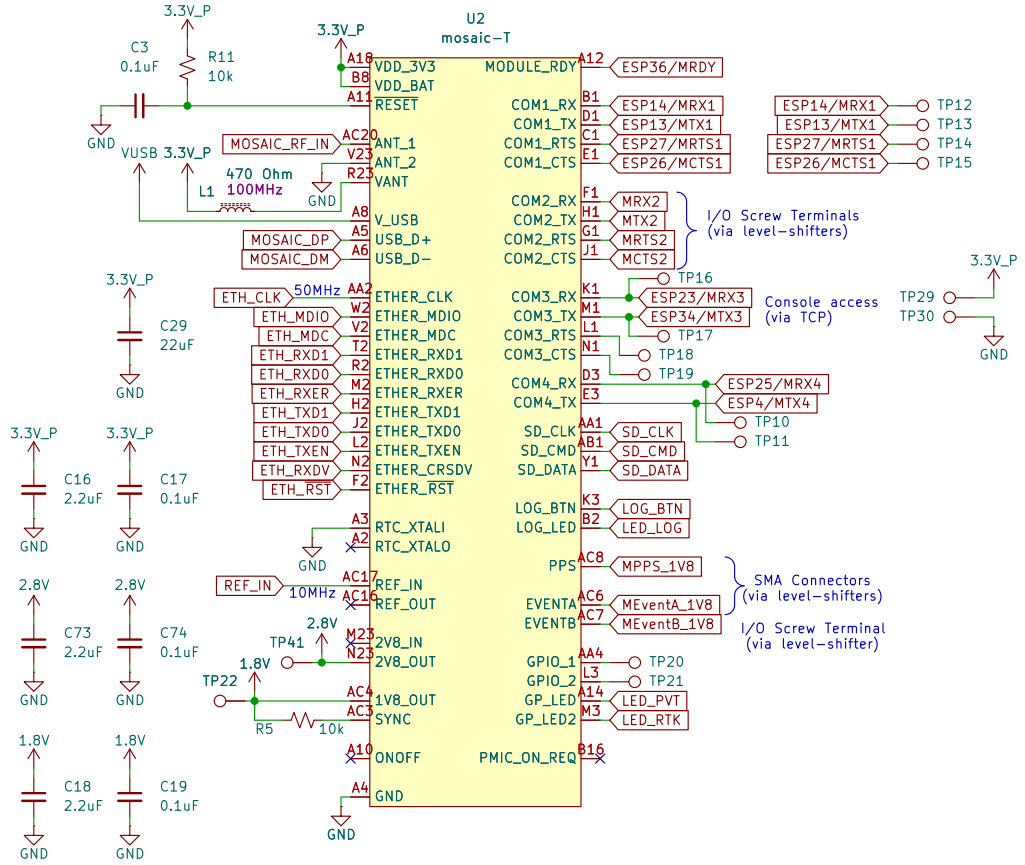


Buttons

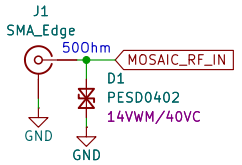


Sheet: /USB/	
File: USB.kicad_sch	
Title: USB	
Size: USLetter	Date:
KiCad E.D.A. 10.0.1	Id: 3/7

mosaic Tri-band GNSS

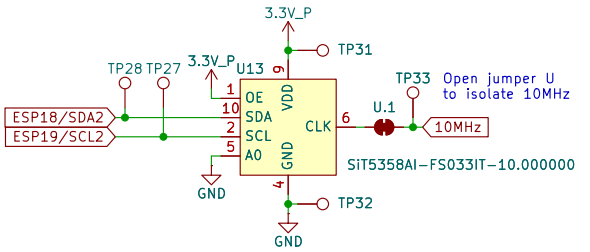


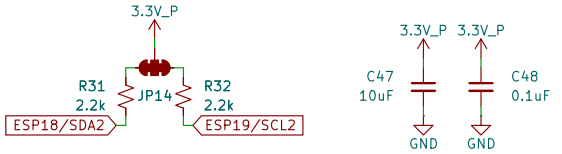
GNSS Antenna



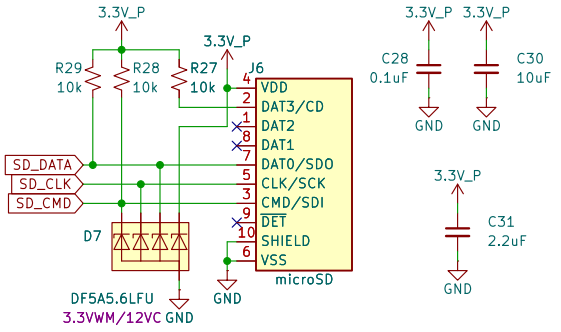
Microstrip Calculation:
Copper Thickness (1oz): 1.4mil/0.035mm
Board thickness: 1.6mm
Dielectric thickness (layer 1 to 2): 0.2mm
Er: 4.6
Polygon Isolation: 6mil/0.1524mm
RF Trace Width: 13mil/0.33mm
https://chemandy.com/calculators/coplanar-waveguide-with-ground-calculator.htm

10MHz Oscillator





microSD

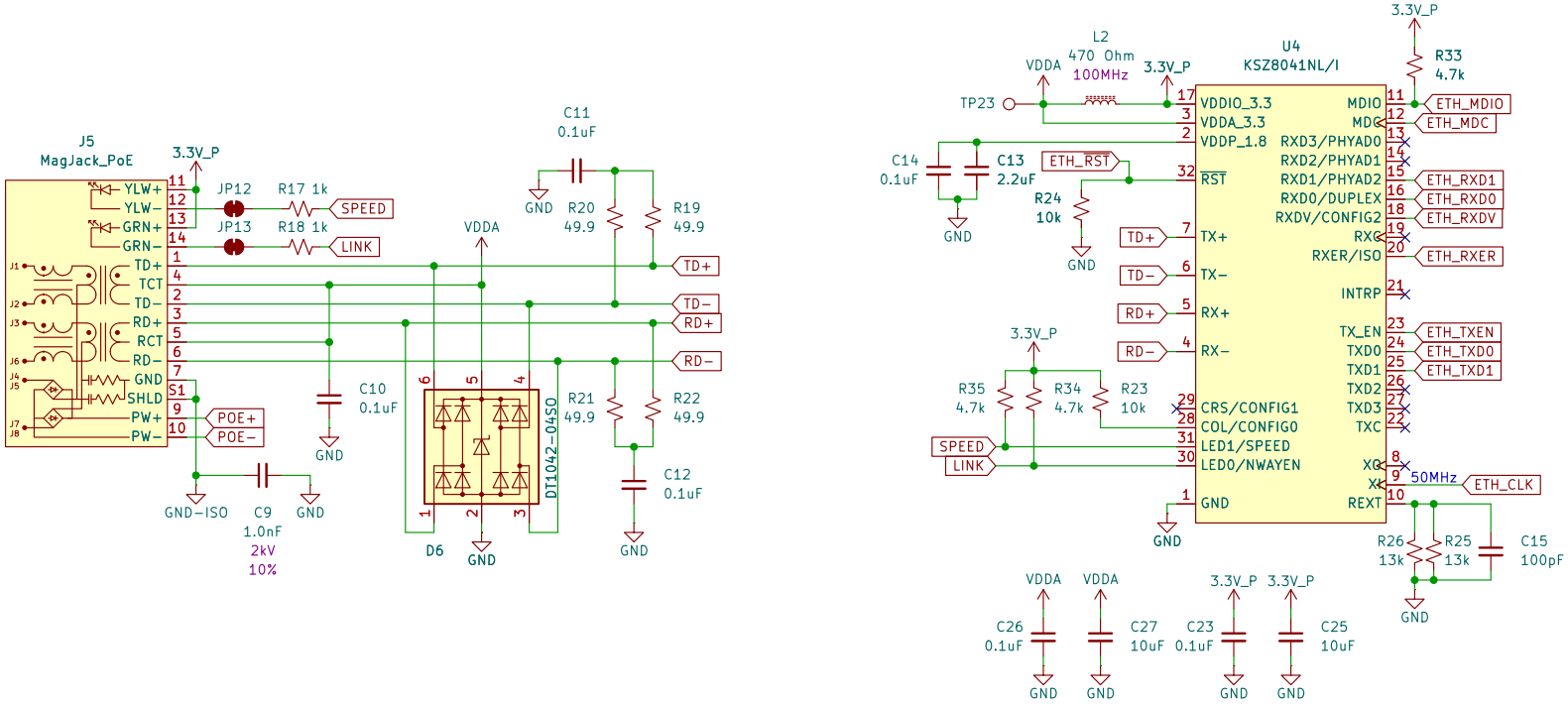


Sheet: /GNSS/
File: GNSS.kicad_sch

Title: GNSS

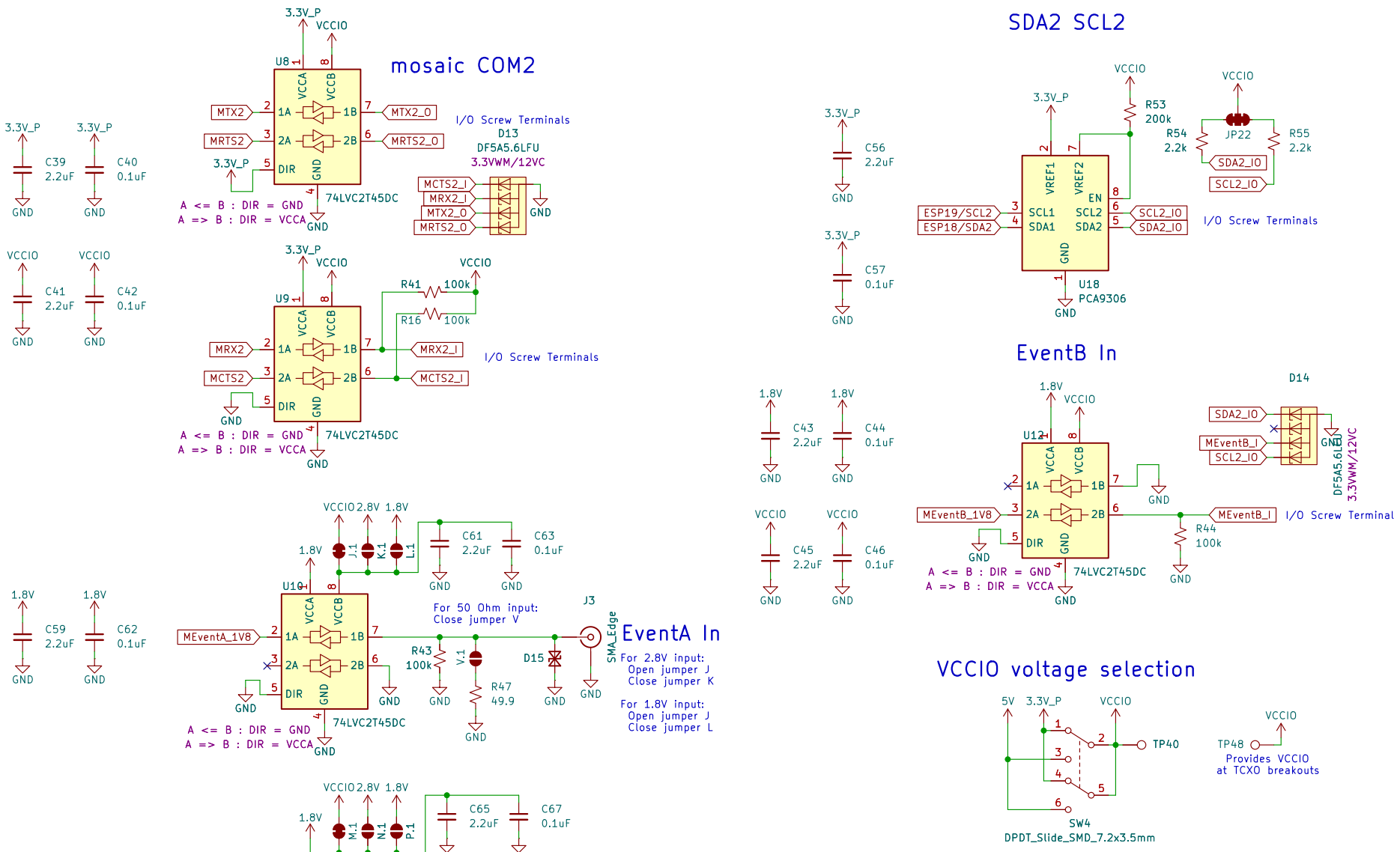
Size: USLetter	Date:	Rev:
KiCad E.D.A. 10.0.1		Id: 4/7

Ethernet



Ethernet Track Impedance: Differential Pair
https://saturnpcb.com/saturn-pcb-toolkit/
Prepreg thickness: 8.3 mil (JLC7628). Er = 4.6
9.0 mil track with 11.0 mil gap (20 mil center to center) = 100 Ohms
Each pair should match in length to better than 0.5mm

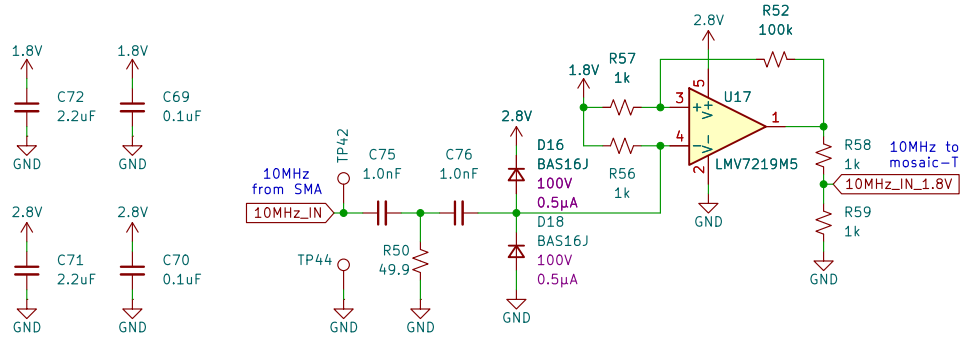
Sheet: /Ethernet/		
File: Ethernet.kicad_sch		
Title: Ethernet		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 5/7



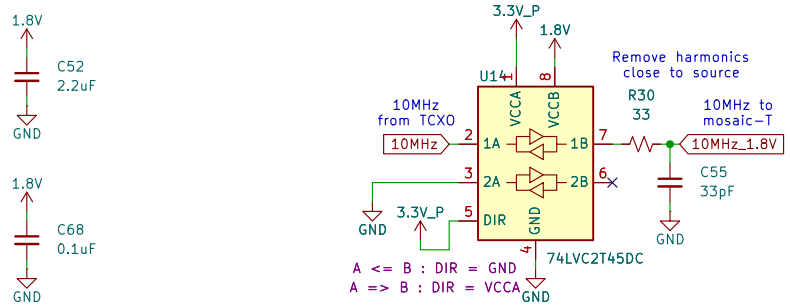
Level-Shifting

Sheet: /Level_Shifting/	
File: Level_Shifting.kicad_sch	
Title: Level Shifting	
Size: USLetter Date:	**Rev:**
KiCad E.D.A. 10.0.1	Id: 6/7

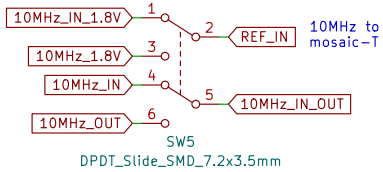
10MHz In
Input impedance: 50Ω
Detection level: −14dBm
Max supported level: +12dBm

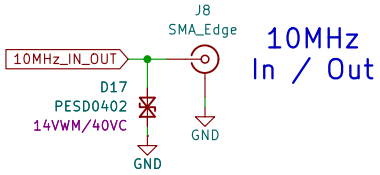


10MHz 1.8V for mosaic−T

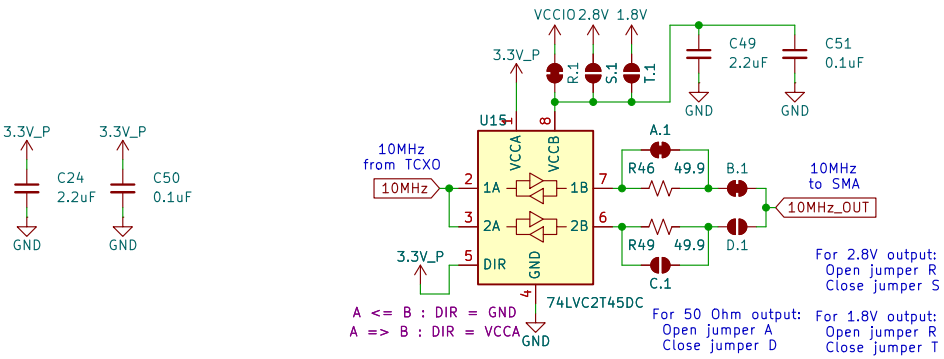


10MHz In / Out





10MHz Out



Level−Shifting 10MHz

Sheet: /Level_Shifting_10MHz/
File: Level_Shifting_10MHz.kicad_sch

Title: Level Shifting 10MHz

Size: USLetter | Date: | **Rev:**
KiCad E.D.A. 10.0.1 | Id: 7/7